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Leading the Way in CCS Monitoring: An Innovative CO₂ Measurement in Reservoirs

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Abstract

An innovative CO₂ measurement in reservoirs utilizes both saturation log and production log tools. The pulsed-neutron water flow log (WFL) survey determined the CO₂ injection profile by monitoring the injection rate and wellhead pressure. The WFL survey was recorded during CO₂ injection at a rate of 4.8 tons per hour at a wellhead pressure of more than 1000 psi. The survey covered the full perforation interval, with WFL stations taken at multiple depths.

The pulsed-neutron tool was logged in GSH (Gas, Sigma and Hydrogen-Index) mode across the injected formation and the shallower seal, covering both ‘Top Seal’ and ‘Storage Formation’ zones. The pulsed-neutron porosity measurement was processed using a new algorithm that compensates for borehole-filled CO₂ gas (as opposed to the conventional algorithm designed for water/oil/gas-filled borehole).

The density measurements in [Fig.1](#) revealed that the water level during injection was pushed down to B ft, while the shut-in water level was at A ft. The minimal injection rate computed below B ft is due to low velocity and high-water holdup. The phenomena and interpretation results showed that the CO₂ injection profile was determined using gas velocity measurements from WFL stations and fluid density measurements.

The presence of gas was indicated by using new algorithm technique and high values of the gas ratio indicator (GRAT) and the difference between pulsed-neutron porosity (TPHI) and effective porosity (PHIE). The lower part of the perforated interval exhibited minimal injection (<5%).

The study concludes that 5% of fraction of the perforation interval of the injected CO₂ went into reservoir, with minimal injection across the lower part. The interpretation results provide valuable insights into the CO₂ injection profile and the effectiveness of the injection process in the first CCS well in Onshore Abu Dhabi.

The evaluation marks a groundbreaking initiative in the monitoring of CO₂ storage within an onshore field in Abu Dhabi. The innovative technique is set to revolutionize both CO₂ injection and monitoring processes and pave the way for better and more efficient CCS wells design/execution.

Introduction

Carbon Capture and Storage (CCS) is a critical technology in the effort to reduce greenhouse gas emissions, involving the capture of CO₂ from industrial sources and its injection into geological formations for long-term storage. Monitoring the behavior and distribution of the injected CO₂ in the subsurface is essential to ensure the effectiveness and safety of the storage process (Xu, et. al.). Traditionally, tracking CO₂ saturation in a reservoir has been challenging due to the unique properties of CO₂ (e.g., zero hydrogen content in the case of supercritical or gaseous CO₂) which can make standard logging measurements less straightforward (Karim et. al., 2024). In Abu Dhabi, the first onshore CCS well was recently completed as part of a pioneering project to permanently sequester CO₂ in a deep saline aquifer, marking a major milestone in UAE's net-zero emission initiatives. A key component of this project is deploying advanced monitoring techniques to evaluate how injected CO₂ is distributed within the reservoir and to verify that the injection is proceeding as intended.

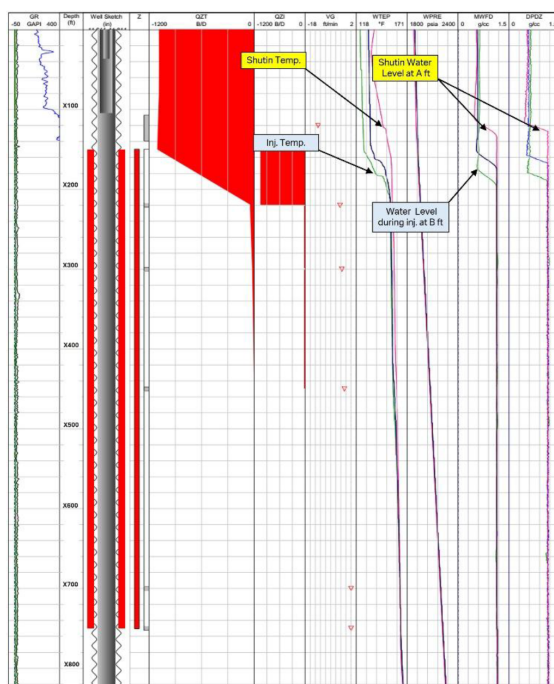


Figure 1—Fluid density (or holdup) log during injection and after shut-in indicates the denser fluid (water) extending to a greater depth during injection (curve shifted downward) and rising after injection stops (curve shifts upward). The depth difference (A to B) represents the displacement of the water column by the injected CO₂ in the wellbore.

This paper presents a comprehensive study of an innovative CO₂ measurement approach in reservoirs, utilizing a combination of saturation logging and production logging tools to monitor CO₂ injection in real-time. Specifically, a pulsed-neutron Water Flow Log (WFL) tool was used in conjunction with new data interpretation algorithms to determine the CO₂ injection profile along the well's perforated interval. The study was conducted on the first CCS injection well in onshore Abu Dhabi, providing valuable insights into CO₂ distribution within the target formation and the integrity of the overlying seal. By integrating a saturation log (pulsed-neutron log) with a production log (WFL), the team could monitor both the saturation changes due to CO₂ and the flow dynamics of fluids in the wellbore simultaneously. This dual-logging strategy is highly novel in the context of CCS monitoring and has the potential to significantly enhance our understanding of CO₂ plume behavior during injection.

Objectives of the Study

The primary objectives guiding this work were: (1) to develop and deploy a logging methodology capable of directly measuring CO₂ saturation and flow in an injection well, (2) to obtain a detailed CO₂ injection profile along the reservoir interval during active injection operations, (3) to evaluate the vertical distribution of injected CO₂ and identify which portions of the perforated reservoir interval are accepting CO₂ and (4) to demonstrate the effectiveness of this monitoring approach for application in future CCS projects. In pursuing these goals, the study also aimed to validate a new interpretation algorithm tailored for CO₂-filled borehole conditions and to document any operational challenges or insights gained from this first-of-its-kind field deployment.

Background and Key Concepts

It is worthwhile clarifying several key terms and concepts relevant to this study. Table 1 provides definitions for important terms used throughout the manuscript, ensuring a common understanding for all readers, including those who may not be specialists in well logging or CCS (Sakurai, et.al., 2006 and Zhou et. al., 2016).

Table 1—Key terms and definitions related to CO₂ injection monitoring and logging techniques used in this study

Term	Definition
CCS (Carbon Capture and Storage)	A technology and process for capturing CO ₂ emissions from sources like power plants or industrial processes and storing it underground in geological formations to prevent it from entering the atmosphere.
WFL (Water Flow Log)	A type of production log used to measure the movement (flow rate and direction) of fluids (especially water) within or behind the wellbore. In this study, a pulsed-neutron based WFL tool is used to monitor injected CO ₂ flow profiles.
Saturation Logging Tool	A well logging instrument that emits high-energy neutrons into the formation and detects the resulting gamma rays and thermal neutrons. It provides information on formation properties like porosity and fluid saturations behind casing by analyzing neutron interactions (capture, scattering) in the formation.
GSH Mode	A specific operational mode of the pulsed-neutron tool that simultaneously measures Gas indicators, Sigma (neutron capture cross-section), and Hydrogen Index (related to porosity and hydrogen content). GSH mode is particularly useful for identifying gas versus liquid saturations in the formation.
CO₂ Injection Profile	The distribution of injected CO ₂ along the reservoir interval – essentially showing how much CO ₂ is entering at different depths or zones of the perforated interval. A profile helps identify zones taking most of the injected gas versus zones taking little to none.
“TOP SEAL” and “STORAGE FORMATION” Zones	Specific stratigraphic zones in the reservoir interval (as referenced in this study). “TOP SEAL” and “STORAGE FORMATION” are acronyms for the reservoir layers under investigation. The pulsed-neutron logging in GSH mode was conducted across both the injection formation (“TOP SEAL” zone) and the overlying seal formation (“STORAGE FORMATION” zone) to assess CO ₂ presence and movement in both the storage reservoir and the caprock above.
GRAT (Gas Ratio Indicator)	A diagnostic indicator derived from pulsed-neutron log data that signals the presence of gas in the formation. High values of GRAT typically suggest a high gas saturation. In this study, GRAT values were used to infer the presence of injected CO ₂ in the reservoir rock.
FNXS (Fast Neutron capture Cross section)	By simple definition, this can be described as the probability of a 14 MeV fast neutron to interact with matter (Zhou et al, 2016). In principle, the FNXS is obtained from GRAT measurement after applying the borehole compensation processing. FNXS value for supercritical CO ₂ (0.6 gr/cm ³) is 2.24 compared with other substances (fluid/rocks) ranging between 7 to 8, signifying its value in differentiating CO ₂ to other non-gas fluids.
TPHI (Pulsed-Neutron Porosity)	The formation porosity as measured by the pulsed-neutron tool (Thermal Porosity, often denoted as TPHI). This measurement can be affected by the types of fluids in the pore space mainly because of their hydrogen index values.
PHIE (Effective Porosity)	The true effective porosity of the formation (often obtained from independent logs or core analysis under fully water-saturated conditions). It represents the fraction of the rock that can store fluids, excluding isolated pores. In this context, comparing TPHI to PHIE helps identify gas: when TPHI is significantly lower than PHIE, it implies that the pore space is not filled with water (or liquid) – consistent with gas (CO ₂) occupying part of the pores.

This project leverages the above concepts in an integrated fashion. Essentially, by using a saturation logging tool in GSH mode in tandem with a Water Flow Log, the team can acquire both saturation data (where is the CO₂ in the formation?) and flow data (how is the CO₂ moving or how much entering each zone?) in one logging operation. The development of a specialized algorithm to process pulsed-neutron porosity in the presence of CO₂ was crucial, as conventional algorithms (geared towards interpreting hydrocarbons or water in the borehole) would produce inaccurate results under these unique conditions. The following sections describe how the logging was executed and the insights gained from the data.

Methodology

Overview

The field experiment was conducted in an onshore Abu Dhabi CCS injection well during an active CO₂ injection operation. The approach involved running a suite of logging tools capable of measuring flow and saturation behind casing without interrupting the injection (Laronga et. al. 2023). The main components of the logging program included a pulsed-neutron logging tool operated in GSH mode and a Production logging tool performing a Water Flow Log (WFL). These tools were conveyed across the well's perforated interval while CO₂ was being injected at steady conditions. Critical injection parameters (rate and pressure) were maintained and recorded concurrently.

WFL Survey Execution

The Water Flow Log (WFL) survey was central to determining the injection profile. This survey was carried out during CO₂ injection, meaning the well was not shut in or killed for logging, ensuring the data reflected actual injection conditions rather than a post-injection state. Key details of the WFL survey are:

- **Injection Conditions:** CO₂ was injected at a surface rate of 4.8 tons per hour (approximately 115 tons per day) throughout the logging operation. The wellhead pressure was maintained above 1000 psi during injection, indicating that the CO₂ was likely in a dense phase (supercritical or near supercritical) as it entered the reservoir. These conditions ensured a continuous stream of CO₂ into the formation, which the logging tools could monitor. Maintaining stable injection pressure and rate was important for data quality, as it provided a steady-state scenario for interpreting flow profiles.
- **Logging Depth Interval:** The WFL survey covered the full perforation interval of the well. In practice, this means that the logging tools started above the uppermost perforation and record data continuously to below the lowest perforation. By doing so, any inflow of CO₂ into the formation along that interval could be detected. The perforated interval corresponds to the open sections of casing where the reservoir was accessed – spanning possibly one or multiple reservoir zones (in this case referred to as the ‘Top Seal’ and ‘Storage Formation’ zones).
- **Station Measurements:** During the WFL, the tool was stationed at multiple depths (logging stations) within the perforated interval. At each station, measurements were taken of fluid flow velocity and possibly fluid holdup. In a CO₂ injection context, a stationary measurement can capture the velocity of fluids inside the wellbore, which helps infer how much flow is leaving the well between stations. For example, if the flow rate measured in the wellbore above a certain depth is higher than below that depth, the difference corresponds to fluid leaving into the formation at that interval. By taking station measurements at fine depth increments, the team could construct a detailed injection profile indicating where the CO₂ was exiting the well and entering the reservoir.
- **Flow and Density Logging:** Modern production logging suites often include not just a spinner (for flow velocity) but also fluid density and pressure sensors. In this survey, fluid density measurements were captured along with flow. The fluid density log in an injection well can reveal

the interface between injected CO₂ and any denser fluid in the wellbore (such as water). In this study, the density log was used to identify the water level in the wellbore during injection, as CO₂ (being less dense) will tend to occupy the upper part of the wellbore and push the water column downward. By comparing density logs during active injections versus after shut-in, the change in water level could be observed.

- **Data Recording and Acquisition:** The WFL data were recorded continuously during the logging run. The log presumably includes a spinner flowmeter reading (to detect fluid velocity in the pipe), a gradiomanometer or density tool (to measure fluid density variation with depth) and possibly an oxygen activation log (typical for water flow detection behind pipe, though in this case we are more interested in in-well flow). The combination of these measurements yields both the rate of CO₂ flow at various depths and the identification of fluid type (CO₂ gas or liquid or both vs water) present at those depths. All tools were calibrated and tested prior to deployment to ensure reliable measurements under the high-pressure CO₂ environment.

Pulsed-Neutron Logging in GSH Mode

Simultaneous with the WFL, a pulsed-neutron log was run to measure formation saturations. The pulsed-neutron tool was operated in GSH mode (Gas–Sigma–Hydrogen), which is specifically suited for CO₂ monitoring due to the following capabilities:

- **Triple-Measurement Mode:** In GSH mode, each logging pass provides three critical pieces of information: Gas indicator, Sigma (Σ), and Thermal Neutron Porosity (related to Hydrogen Index).
 - **The Gas indicator** (GRAT / FNXS) is sensitive to the presence of gas, both in the borehole as well as in the formation. The stronger the gas presence is, the higher GRAT value will read and FNXS will be lower. (Note that due to the nature of fast neutrons, the depth of investigation of this measurement is quite shallow; depending on the logging environment or well completions, it may primarily responds to annulus (or annuli) fluid rather than formation).
 - **The Sigma (Σ) measurement** is the macroscopic thermal neutron capture cross-section of the formation. Sigma is influenced by the fluids in the pores because different fluids have different capture characteristics (for instance, saltwater has a high sigma due to dissolved salts like Cl⁻, whereas hydrocarbons and CO₂ tend to have low sigma). Monitoring sigma can thus help infer changes in water saturation; if CO₂ replaces brine, the sigma should drop in those intervals.
 - **The Hydrogen Index / Neutron Porosity (TPHI) measurement** provides a porosity reading based on hydrogen content. It essentially indicates how much hydrogen is in the formation's pore space, which correlates to fluid-filled porosity if the fluids contain hydrogen (water or oil have hydrogen, CO₂ does not). A drop in neutron porosity (TPHI) compared to the original water-filled porosity (PHIE) can be diagnostic of gas replacing liquid in the pores.
- **Depth Coverage ('Storage Formation' and 'Top Seal'):** The pulsed-neutron tool was logged across both the injection zone and the seal zone, specifically covering the 'TOP SEAL' and 'STORAGE FORMATION' zones mentioned earlier. The 'STORAGE FORMATION' zone corresponds to the reservoir interval where CO₂ is injected, and the 'TOP SEAL' zone is an overlying formation. Logging through the seal (caprock) provides a baseline to ensure that CO₂ has not breached the storage formation. It also gives context to the reservoir readings by providing measurements in a zone that should remain at original condition.
- **Logging Procedure:** The pulsed neutron logging was performed in the cased-hole environment, through several different layers of tubing/casing (single up to triple layers of barrier). Standard practice for pulsed-neutron logging involves either station measurements or a continuous log. Given that WFL stations were taken, and at those same station depths the pulsed-neutron tool

also recorded stationary measurements to get a stable reading of sigma and porosity. This allows for more precise capture of formation properties under injection conditions. It's also possible a continuous log was recorded on an additional pass. The critical requirement was timing the logging such that CO₂ injection effects (like pressure and saturation) were stable.

- **Data Processing Onsite:** Immediately after acquiring the pulsed-neutron data, preliminary processing was carried out using a new algorithm for porosity interpretation (described in the next subsection). Quick-look analysis at the wellsite focused on identifying zones with anomalously low TPHI (indicating gas) and high GRAT values. This preliminary interpretation confirms whether CO₂ was indeed detected in the intended injection zone and not into the caprock. It would also reveal qualitatively how CO₂ is distributed (e.g., more gas effect in the top perforations or bottom perforations).

New Porosity Interpretation Algorithm for CO₂-Filled Borehole

One critical technical challenge addressed in this study was how to correctly interpret thermal neutron porosity measurement when the borehole is filled with CO₂ gas. Originally, the pulsed-neutron porosity algorithm was designed and optimized for certain conditions where the borehole is filled with either water, oil, or hydrocarbon-gas along with varying completion types. With experiences of logging the pulsed neutron tool in CO₂-filled boreholes, it became apparent that the standard TPHI algorithm was not accurate in such condition due to the extreme difference in neutron transport properties between CO₂ and the more common borehole fluids. In 2024, a dedicated CO₂-filled borehole TPHI algorithm started to be developed based on Monte Carlo N-Particle (MCNP) modeling. The algorithm is based on the difference in the compensated ratio of the Near-Far and Near-Deep detectors, and it applies additional correction to the standard TPHI calculation. This novel algorithm is expected to be fully commercialized by the end of 2025 or early 2026.

Several key features of this algorithm include:

- **Borehole Correction:** Instead of using the standard borehole correction (as described above), the algorithm inputs the properties of CO₂ (at the downhole pressure/temperature) as the borehole environment. CO₂ at high pressure is less dense than water and has different neutron cross-section measurement and sigma formation properties but it is highly sensitive to variation in gas volume comparing with in water volume. The algorithm adjusts the neutron porosity (TPHI) calculation so that the absence of hydrogen in the borehole does not erroneously make the formation appear less porous or more gas-saturated than it actually is. In effect, it recalibrates the tool's response for a CO₂-filled completion.
- **Dynamic Range for Gas Detection:** The new processing enhances the dynamic range for detecting gas in the formation. When the borehole itself is filled with gas, distinguishing formation gas from borehole gas can be tricky. The new processing provides more reliable values of TPHI in the presence of borehole CO₂. The **difference between TPHI and PHIE** is a crucial output: **under proper calibration, a large TPHI deficit (TPHI << PHIE)** in a zone indicates that injected CO₂ has displaced what was originally formation water. The algorithm ensures this difference is calculated accurately by not attributing the low hydrogen in the wellbore to the formation.
- **Validation:** The algorithm has been tested on synthetic data and is currently being field tested. In practice, one might validate it by looking at known 100% water zones (like the caprock above injection) – there, even with CO₂ in the borehole, the algorithm should output TPHI approximately equal to PHIE (since formation is all water). That was likely confirmed in the 'STORAGE FORMATION' zone logging. Another validation is internal consistency: for example, if a zone is known to have low original gas (as per pre-injection logs or core data), any gas indication now

should be due to injected CO₂, and the magnitude should align with the injected volume and flow profile.

- **Conventional vs. New Algorithm:** It is worth noting how the new algorithm differs from conventional ones. Conventionally, cased-hole neutron porosity tools use a calibration where the borehole effect is a relatively minor correction if filled with water (because water has a high hydrogen index, similar to formation fluids). With a gas-filled borehole, conventional interpretation might mistakenly register a very low porosity or even negative porosity anomaly, as the tool primarily sees much fewer hydrogen atoms (making even a high-porosity formation appear low-porosity). The new algorithm avoids such misinterpretation, effectively enabling **quantitative CO₂ saturation monitoring** rather than just qualitative detection.
- Having described the tools and procedures, the next section will present the results of the logging survey, including the CO₂ injection profile obtained and the observations made about CO₂ distribution in the formation.

Result

The results include quantification of injection rates at various depths, identification of the CO₂ entry points into the formation, movement of fluids in the wellbore (water vs CO₂), and detection of CO₂ within the reservoir rock pore space. Key numeric results are highlighted in the metrics below, and detailed explanations follow:

Injection Conditions and Wellbore Fluid Movement

Stable Injection Regime. Throughout the logging operation, the surface injection rate was held at 4.8 tons/hour and the wellhead pressure remained above 1000 psi. These steady conditions meant that the wellbore was filled predominantly with CO₂ from the injection point up to a certain height, above which any lighter fluid (gas) would accumulate. Indeed, the fluid density log acquired during injection clearly showed a transition from dense fluid to lighter fluid at a particular depth. This transition corresponds to the water level in the well. The density measurement revealed that the water level was pushed down to a depth denoted as B (ft) during injection, whereas after the well was shut in, the water level rose back up to depth A (ft). In other words, active injection of CO₂ forced the borehole's liquid (water) column downward by (B – A) feet, displacing it with CO₂ gas. This observation confirms that the wellbore was indeed filled with injected CO₂ down to the lower depth B while injection was ongoing. It also provides a visual gauge of the volume of CO₂ in the wellbore annulus: a deeper water level during injection implies a larger volume of CO₂ present in the casing.

From an operational standpoint, the depression of the water level by CO₂ is expected but measuring it provides a useful piece of data to cross-check CO₂ volumes. It also serves to corroborate with the WFL data. If CO₂ were leaking into an unperforated zone or behind pipe unexpectedly, the density log might show an irregular profile. In this case, the smooth and stable displacement of fluid suggests that CO₂ is behaving as intended in the wellbore, moving downward only as far as the perforations allow it to enter the formation.

CO₂ Injection Profile Along the Perforations

The key result from the WFL survey is the CO₂ injection profile, essentially, how the 4.8 tons/hour of CO₂ is distributed across the perforated interval. By comparing flow measurements at various stations, the logging team deduced the fraction of CO₂ going into different segments of the reservoir. The interpretation revealed a highly uneven distribution:

- **Top vs Bottom Injection:** The 5% of upper part of the perforation interval was injected by CO₂, whereas the 95% of the interval took very little. In numerical terms, the lower portion of the

perforated zone exhibited minimal injection, on the order of less than 5% of the total CO₂ injected. Over 95% of the injection was entering the upper perforations. This observation suggests that some variation in injectivity is present, possibly due to differences in permeability and connectivity behind the pipe between the perforation interval.

- Determining the Profile:** Consider that at the top of the perforated interval, essentially 100% of the injected CO₂ is still in the wellbore (just above the top perforation). As the log tool moves down and passes the first set of perforations, if some CO₂ exits into the formation there, the flow in the wellbore will drop by that amount. The WFL spinner tool would measure a lower flow rate below those perforations. In this survey, the flow difference measured before and after each zone of perforations was used to calculate the CO₂ into that interval. The fluid velocity readings at WFL station depths, combined with knowledge of the tubing/casing cross-sectional area and fluid density, yield volumetric flow rates. The fluid density readings help confirm whether the flowing fluid is CO₂ gas or liquid; in this case, it would be primarily gas, but if any water were moving it could be distinguished.
- Quantification:** The profile data indicated that nearly all the flow reduction occurred across the upper perforations, meaning nearly all CO₂ went into that upper reservoir segment. The total CO₂ injection rate at downhole conditions (considering pressure and temperature) was estimated to be around 1000 barrels per day entering the formation. This figure is consistent with the surface tonnage rate when converted (since CO₂ at reservoir conditions has a density that makes 4.8 t/h roughly equal to 1000 BPD of fluid). It provides a check that the profiling accounted for essentially the entire injected volume (no significant unaccounted losses).

Detection of CO₂ in the Reservoir (Saturation Logging Results in Fig.2)

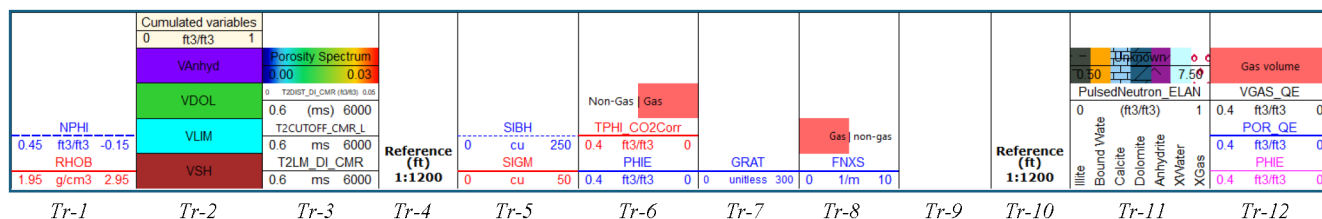


Figure 2—Pulsed-neutron composite display header

While the WFL gave information about where CO₂ is going *into* the formation, the pulsed-neutron tool in GSH mode provided insight into the *state of the fluids inside the formation pores*. The results from the saturation log can be summarized as follows:

Figure 2 below shows the details of the track headers used for subsequent discussions on the pulsed neutron results.

Track-1: OH Neutron - Density

Track-2: OH Lithology volumes

Track-3: OH NMR Data (T2-Distribution, T2LM, and T2 Cutoff)

Track-4: Measured Depth (ft.)

Track-5: Formation Sigma (SIGM) and Borehole Sigma (SIBH)

Track-6: OH Porosity (PHIE) and pulsed-neutron Porosity (TPHI_CO2Corr)

Track-7: Gas Ratio indicator (GRAT)

Track-8: Fast Neutron capture cross-section (FNXS)

Track-9: Completion Diagram

Track-10: Measured Depth (ft.)

Track-11: CH ELAN outputs (litho & fluid volumes)

Track-12: OH Porosity (PHIE) vs CH Porosity (from ELAN, POR_QE) and Gas volume in formation (VGAS_QE)

Figure-3 presents the results from the pulsed neutron-based logs across the perforation intervals. In this display, the Fast Neutron Capture Cross Section (FNXS – Track-8) measurement (derived from the GRAT curve – Track-7) showed a **very strong and clear gas signature in the upper part of the perforated interval**. This is also supported by looking at the deficit of TPHI when compared to OH-PHIE curve (Track-6). In fact, the gas signature can be observed from above X100 ft in this picture, down to ~X195 ft.

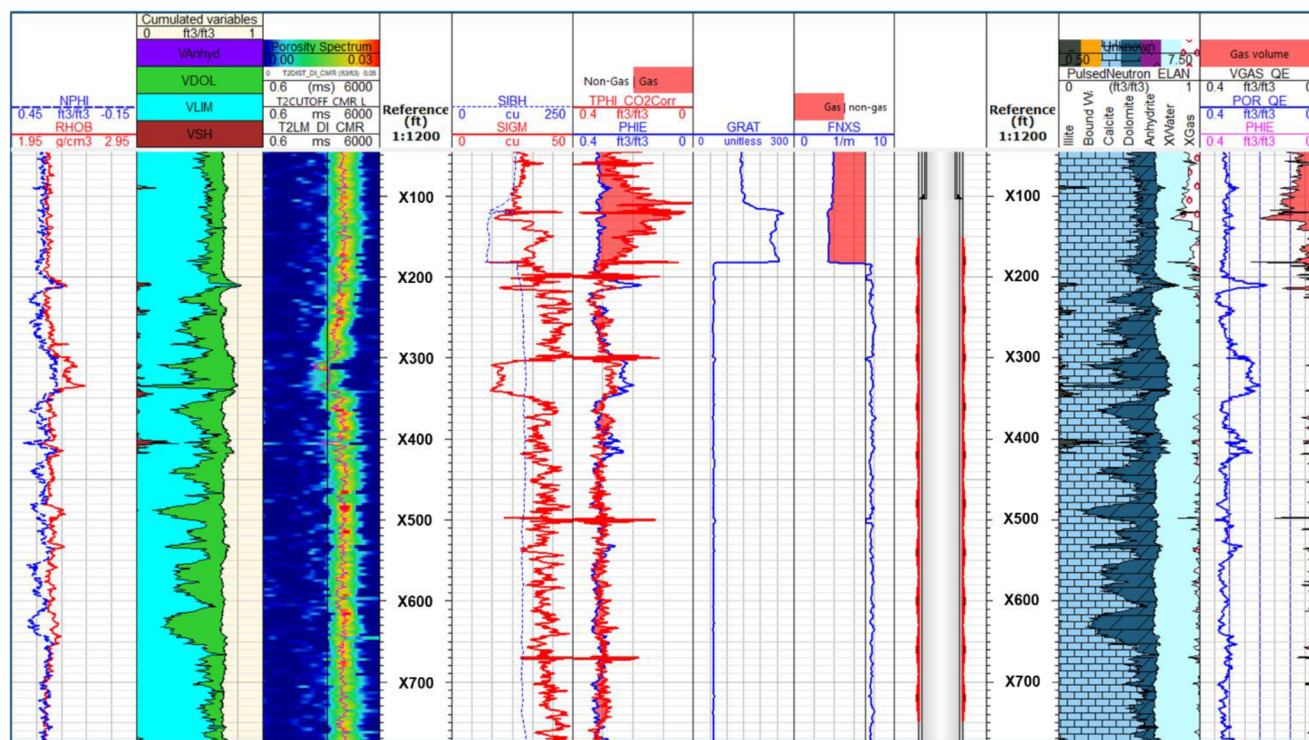


Figure 3—Pulsed-neutron Composite display across Perforation intervals

Interestingly, the intervals below X195 ft. down to the bottom of the perforation zone have shown very little to no gas presence, both in formation and borehole (this is in line with the results obtained from PLT log). Note that the traces of gas volumes computed by ELAN below X195 ft. most likely arise due to measurement/computation uncertainty and should be kept into consideration when looking at this zone.

Figure-4 shows the composite display between the caprock zone down to top of perforation intervals. The clear gas signature presence can be observed from TPHI and FNXS/GRAT between W925 to X195 ft. Most likely, the injected CO₂ has been migrating upwards from its entry zone in the top part of the perforation interval.

Another key piece of information from this plot is that all the indicators (GRAT/FNXS, TPHI deficit, sigma changes) in the caprock (W360 ft. and above) zone **did not** show anomalies. This means the CO₂ was indeed confined to the intended reservoir and had not breached into the caprock. That is a critical result for storage security. If any CO₂ had been detected in the seal, it would have raised concerns about caprock integrity (cement bond log acquired to validate) or unseen pathways; but the data did not indicate such issues.

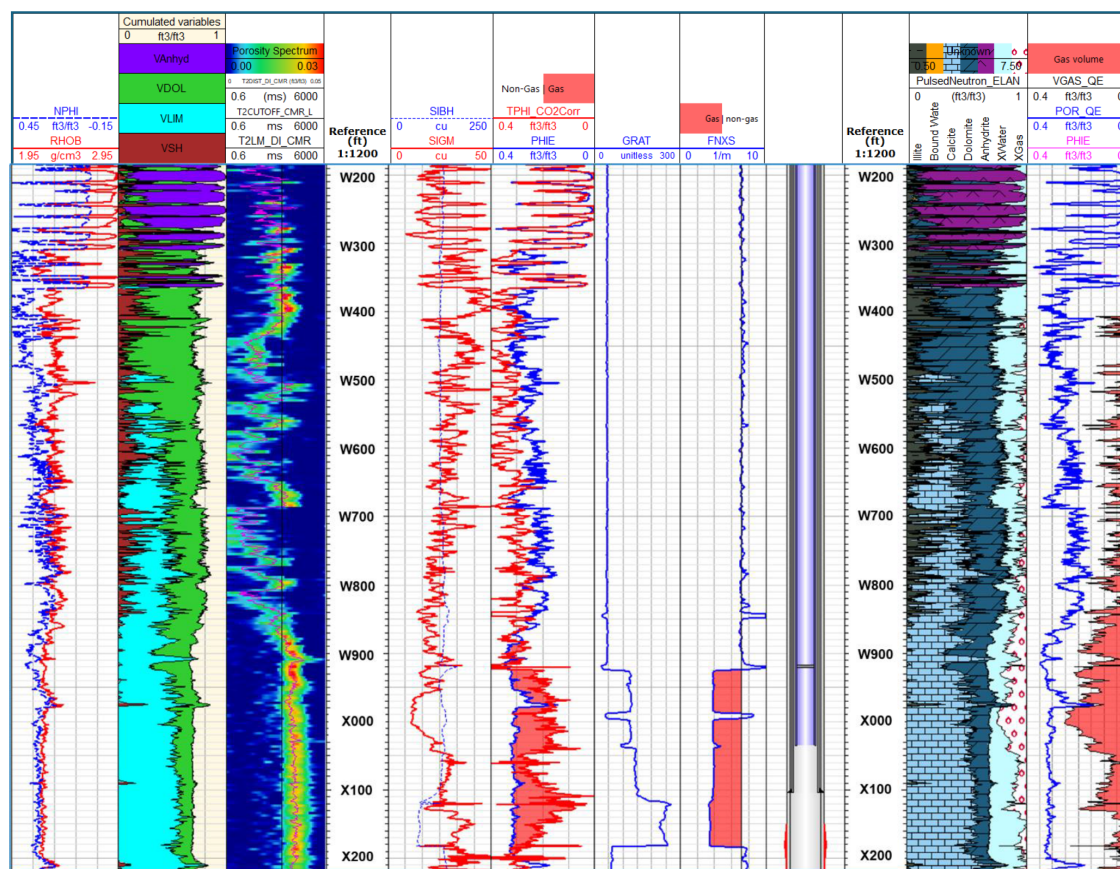


Figure 4—Pulsed-neutron Composite display above Perforation intervals up to Caprock zone

Considering the absence of TPhi-PHIE deficit and FNXS/GRAT reading, the computed gas volumes between the bottom of caprock (W360 ft.) down to W925 ft. carries a high degree of uncertainty. Based on the NMR T2 Distribution reading (Track-3), we can observe a clear change in the T2 distribution reading at around ~W800-W835 ft. with the T2 response becoming faster at this depth compared to the depth below where the gas signature is quite clear. This indicates the possibility of changes in rock pore size and/or grain size distribution, making this zone a natural barrier for the upward gas migration.

Integrated Interpretation – CO₂ Injection Effectiveness

Combining the above results yields a coherent story of the CO₂ injection process in this well. Most of the injected CO₂ penetrated the upper part of the reservoir interval, filling the available pore space there (displacing brine and registering clearly on logs), and only a negligible amount of CO₂ reaching the lower part of the interval. The total injected volume as recorded by surface metering was accounted for by the logging data (about 1000 BPD into the formation), and no leakage or unexpected migration of CO₂ was observed outside the target zone. The first CCS well in onshore Abu Dhabi thus appears to be performing as designed in terms of containment, but with an injection distribution that is skewed towards the top of the perforated formation.

Before moving to the broader implications, it's worth emphasizing how novel these results are. Rarely has a CCS injection well been instrumented with such logs during active injection to directly observe CO₂ behavior. This real-time look provided an invaluable dataset for understanding CO₂ flow in the subsurface. In the following discussion, we interpret what these findings mean for reservoir management and future CCS operations.

Discussion

The results obtained from this innovative CO₂ monitoring approach have several important implications. The discussion is structured around the significance of the findings for the current project and how this knowledge can influence future CCS projects in similar settings.

Effectiveness of the CO₂ Injection Process in the First CCS Well

From a broader perspective, how effective was the injection in meeting its goals? The main goals in a CCS context are to inject CO₂ safely, in large volumes, and have it securely stored. The monitoring results indicate:

- **High Conformance to Containment:** The CO₂ stayed in zone, a big success in terms of storage security. No breach of the caprock was detected and there was no sign of an out-of-zone injection. This is extremely reassuring for a first-of-its-kind project. It validates the site selection and well design choices regarding containment.
- **Injection Volume Achieved:** The target injection rate (4.8 t/h, ~115 t/day) was achieved and maintained. Over the period of the test/logging, this means several tens of tons of CO₂ were injected and all successfully went into the formation. In terms of well injectivity, this indicates the well is capable of handling commercial-scale injection rates.
- **Utilization of Storage Capacity:** The downside is that only part of the intended reservoir thickness is taking the CO₂. The effectiveness in terms of distributing CO₂ is lower than ideal. However, this early monitoring can guide later strategy (for example, injecting slower to let CO₂ redistribute, or adding another perforation in a shallower zone, etc.).
- **First CCS Well Benchmarks:** Given that this is the first CCS well in onshore Abu Dhabi, these results set a benchmark. The data will be a reference point for what injection profiles might look like in similar formations (saline carbonate aquifers). It also demonstrates that such monitoring is feasible and highly valuable. The lessons learned here have already improved the understanding of how CO₂ behaves in this specific geologic setting.

In conclusion, the project was successful in safely emplacing CO₂ into the selected reservoir and the monitoring exercise was successful in diagnosing the process. The well is functioning as a secure injector and we have identified an area for improvement (better distribution across the interval).

Novelty of the Monitoring Technique and Its Impact

The approach of combining WFL and pulsed-neutron logs in an active injection well is a novel technique that proved its worth. This evaluation is described as a groundbreaking initiative in onshore CO₂ storage monitoring:

- **Direct Measurement vs. Indirect Monitoring:** Traditional CCS monitoring often relies on indirect methods like seismic surveys, pressure monitoring, or sampling from observation wells, which provide data at the field-scale but not the fine resolution in the well. This technique, however, gives direct high resolution in-situ measurements of CO₂ in the reservoir at the injection point. Time-lapse seismic might eventually show if CO₂ is spreading unevenly, but the early logging data identified it immediately and quantitatively.
- **Real-Time Decision Making:** The ability to run these logs during injection means operators can get near-real-time feedback. If something were amiss (say CO₂ was going into an unintended layer, or one segment was not taking any CO₂ at all), they could have adjusted operations on the fly. In the context of this test, it was also a demonstration that logging tools can survive and function in

the CO₂ environment, paving the way for routine use of these logs in CCS wells without needing to stop injection.

- **Algorithm and Interpretation Advancements:** The custom algorithm for CO₂ borehole compensation is an innovation that extends the capability of existing logging hardware. This means any similar pulsed-neutron tool can be upgraded via software to handle CO₂ scenarios. The impact is industry-wide; other CCS projects can adopt this approach, applying the algorithm to get accurate readings. This study expanded the toolkit available for subsurface engineers tackling greenhouse gas storage.
- **Comprehensive Data Acquisition:** In one combined log run, the team gathered information on flow, pressure, fluid identification, saturation, porosity and sigma. This is an extremely rich dataset and will encourage more integrated logging runs in the future, moving away from single-purpose logs to multi-faceted logging campaigns especially in critical wells like CCS injectors.

This approach is likely to become a reference model for CCS well monitoring.

Implications for Future CCS Well Design and Execution

Several takeaways from this study can guide future CCS well design and operational execution:

- **Perforation Strategy:** If heterogeneity is anticipated (or once it's observed in a first well), future wells might choose to segment the injection interval. For example, instead of perforating a long interval and hoping for uniform injection, an engineer could install packers to isolate sub-intervals and inject into each separately. This would force CO₂ into each layer in a controlled manner.
- **Logging as Part of Commissioning:** This study makes a strong case that any new CO₂ injection well should be logged in a similar way as part of commissioning. Designing the well completion to allow such logging (e.g., ensuring the well can be logged with wireline under injection conditions, perhaps by having a large enough tubing or memory tool deployment) will be considered in future well plans.
- **Wellbore Material and Safety:** The experience gained in running logging tools under high CO₂, high-pressure conditions will influence how tools and well materials (like packers, completion metals, elastomers) are selected. Future operations will benefit from the lessons learnt on how to handle CO₂ (for instance, managing the risk of CO₂ leaks at surface while logging, or sensor calibration in CO₂).
- **Data Integration:** When planning monitoring for future projects, integrating different data types will be key. For instance, coupling the log results with reservoir simulation models. The ~1000 BPD injection into mainly the upper zone can be history-matched in a reservoir model to predict longer-term CO₂ migration. If only surface data were used, the model might have assumed uniform distribution, which would be wrong. Hence, the modeling and design of CO₂ storage projects will increasingly rely on such high-fidelity measurements.

In summary, the execution of this monitoring program on the first CCS well provided insights that are immediately useful for improving future operations. It underscores the idea that monitoring is not just for safety, but also for optimization.

Challenges and Limitations

It is important to recognize potential challenges encountered and aspects that limit interpretation:

- **Logging Under Injection Conditions:** Running tools in a live injection well can be challenging. In this case, it was done successfully, but generally there are risks such as tool clogging with hydrates (CO₂ can form dry ice if the wellbore cools during logging), or difficulties in log interpretation due

to non-stationary conditions. The data might have had noise due to flow turbulence. Careful station keeping was needed to get good neutron log readings. Success here speaks to good operational execution, but it is a complex operation that requires expertise.

- **Resolution of WFL:** The WFL could reliably detect the big differences in flow (95% vs 5%), but imagine if distribution were more even, small differences might be harder to resolve. The accuracy of spinner flowmeters and density tools in multiphase flow conditions could be a limiting factor. If some CO₂ was dissolving in water or carrying water droplets, the interpretation becomes challenging. In this test, the contrast was stark, which helps confidence.
- **Algorithm Validation:** The new neutron interpretation algorithm is certainly a step forward, but it may need further enhancements. For example, if another independent measure of saturation (like pulsing CO₂ then shutting in and doing a temperature log or using tracers) were available, it could cross-validate the indicated saturations. Without core data under those conditions, the algorithm's results rely on assumptions of neutron physics in CO₂ which, while based on theory, always benefit from field confirmation.
- **Data Interpretation Complexity:** Merging datasets (flow and neutron) required careful integration. There could be depths where signals were ambiguous; for instance, a slight CO₂ entry that's below detection of spinner but shows up in neutron log, or vice versa. Reconciling such discrepancies can be a challenging task, requiring deep expertise.
- **Acknowledging these limitations,** the overall confidence in the results is high because multiple lines of evidence (flow, density, neutron, sigma) all pointed to the same conclusions. Future work would aim to refine the measurements (e.g., even higher-resolution logging tools, or logging multiple times to see if distribution changes).

Conclusion

This study demonstrated a successful deployment of an innovative CO₂ monitoring methodology in a real-world CCS injection well, providing rich insights into the injection dynamics and storage behavior that would be unattainable with conventional monitoring alone. By utilizing a pulsed-neutron log in GSH mode alongside a water-flow log, the team was able to determine the CO₂ injection profile with depth, directly observe CO₂ presence in the reservoir, and verify the integrity of the storage formation in real-time.

Key conclusions include:

- (1) The new logging interpretation algorithm successfully adjusted for a CO₂-filled borehole environment, enabling accurate differentiation between formation gas saturation and borehole effects, which proved crucial for correct analysis of the log data;
- (2) The CO₂ was effectively contained within the intended zone. No leakage into the overlying formation was detected, affirming the caprock's sealing capacity and the overall security of storage;
- (3) 95% of the injected CO₂ is entering the 5% of the upper perforation interval, with minimal CO₂ reaching the lower portion (<5% of total) indicating strong vertical heterogeneity in injectivity;
- (4) This integrated logging and its application in the first onshore Abu Dhabi CCS well, sets a precedent for future wells in the region and beyond.

In summary, the evaluation carried out in this study marks a groundbreaking step in how we monitor and understand CO₂ behavior underground. The techniques and knowledge gained will pave the way for better and more efficient CCS well design and execution in the future, ultimately contributing to safer and more effective carbon sequestration operations worldwide. The success of this project encourages wider adoption of advanced downhole monitoring to ensure that as we scale up CCS, we do so with the clarity and confidence that these tools and techniques can provide.

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